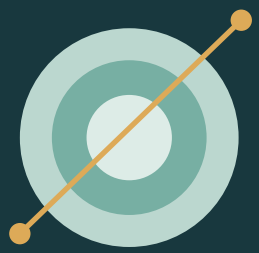
Lumen

Source-linked support for pathology-report conversations

PROJECT PURPOSE

Keep the source visible.
Keep the clinician in the loop.

Lumen explores a focused way to make report wording easier to discuss by tying statements to exact source text and offering neutral questions - without making care decisions.



SYNTHETIC ONLY • ENGLISH ONLY • MODEL OFF

8	46	34 / 34	64 + 10
FICTIONAL REPORTS	**EXPECTED FACTS**	**GOLD SPANS**	**SOFTWARE TESTS**
Frozen synthetic set	Hand-authored labels	Exact quote + offset	Backend + browser

WHAT THE PROJECT IS

An English-only, text-only prototype for discussing pathology-report wording. Its interface emphasizes exact source quotations, explicit **not stated** fields, and neutral care-team questions.

WHAT IT DOES NOT DO

It does not diagnose, predict prognosis, recommend or rank treatment, advise doses, decide trial eligibility, or triage urgent symptoms. No patient benefit or clinical outcome has been measured.

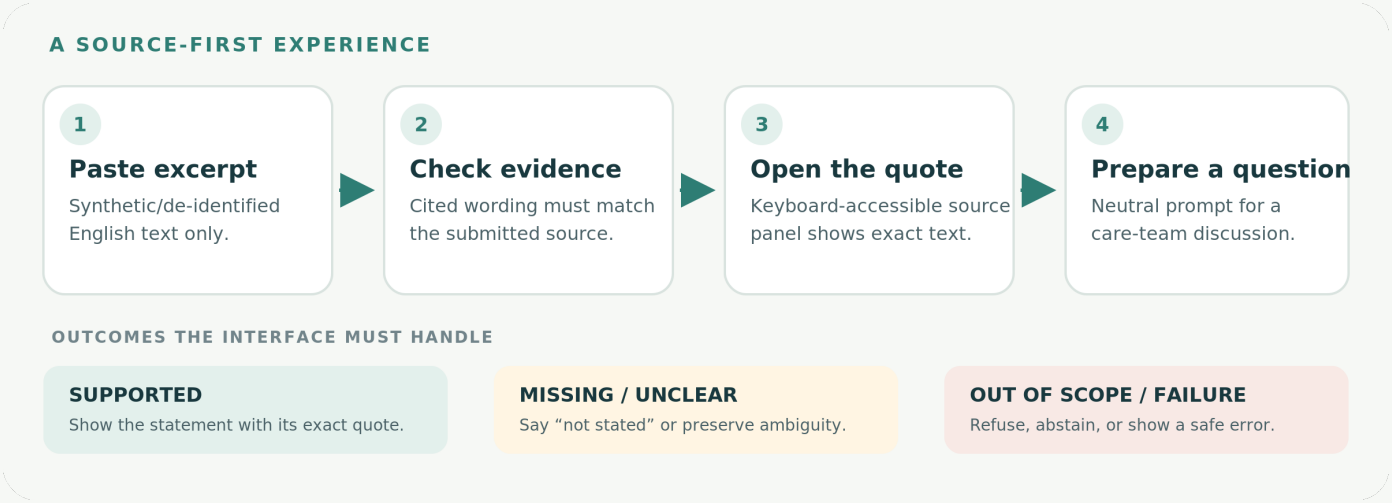
CURRENT PROJECT STATUS

The model provider is disabled. The live browser-to-service path returns a generic unavailable response with no claims. The positive result screen is a synthetic UI test fixture only - not model output.

Read this report as: software behavior and synthetic-data integrity evidence, not model accuracy or clinical validation.

A source-first patient experience

The interaction is designed to move from a short excerpt to a grounded discussion - and to stop safely when evidence is missing or a request is out of scope.



SYNTHETIC EXAMPLE • SOURCE TEXT ONLY

Lung, right upper lobe, biopsy: Adenocarcinoma, acinar pattern.

Exact fictional source wording; shown to illustrate citation mapping, not a model-generated explanation.

WHAT A USER CAN SEE

Before analysis - text-only helper, character limit, explicit synthetic/de-identification acknowledgement, and disclosure that a configured provider may process text.

In a supported result - claim-specific keyboard control, exact labeled source quote, and neutral care-team question.

When information is absent - the interface says “not stated in this excerpt,” not “negative” or “normal.”

SAFE STOP STATES

Unsupported statement - omitted by verification; rejected text is not rendered.

Out-of-scope question - fixed refusal; no report claims are shown.

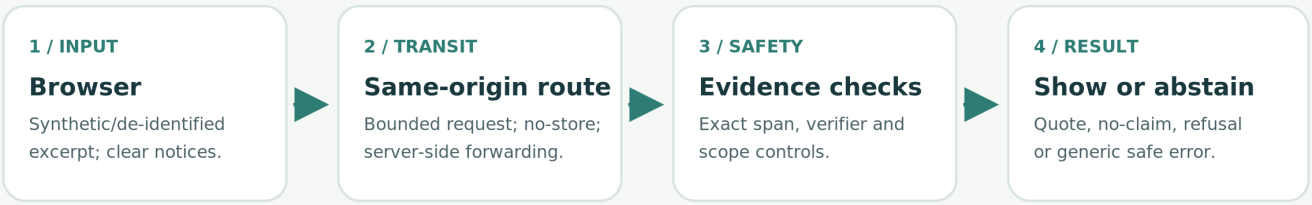
No claims or service outage - calm generic message; stale claims are cleared.

No upload, PDF/photo/OCR, translation, trial search, or report persistence in URL/browser storage is part of this prototype.

Safety and privacy are product boundaries

The data path is intentionally narrow. These are software controls and disclosures; they do not establish clinical safety, privacy compliance, or live-model performance.

HIGH-LEVEL DATA FLOW AND PRIVACY BOUNDARY

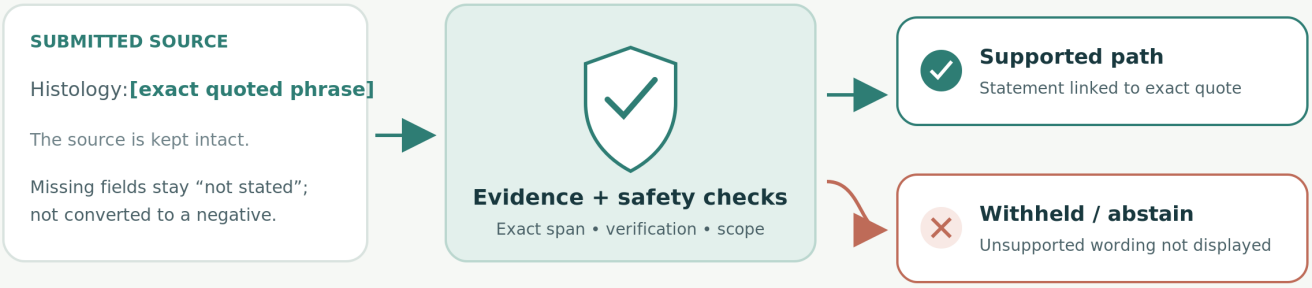


Current demo: provider disabled

The live browser-to-service path returns a generic unavailable response and no claims.
If enabled later, text may leave the device; provider retention, region and terms need review.

HOW THE EVIDENCE GATE IS DESIGNED

Software design; live model output not evaluated



SAFETY MECHANISMS

1. **Exact evidence** - citation quote and offsets checked against submitted text.

2. **Constrained structure** - strict fields limit output shape.

3. **Separate verification** - the intended pipeline has distinct generator/verifier roles.

4. **Fail closed** - refusal, abstention, generic error or withheld claim when checks fail.

PRIVACY BOUNDARY

Current demo: provider disabled; no external model request was made. The app has no report database and does not place report text in URL or browser storage.

If enabled later: submitted text may leave the device for the operator-configured provider. Retention, region and terms need deployment-specific review.

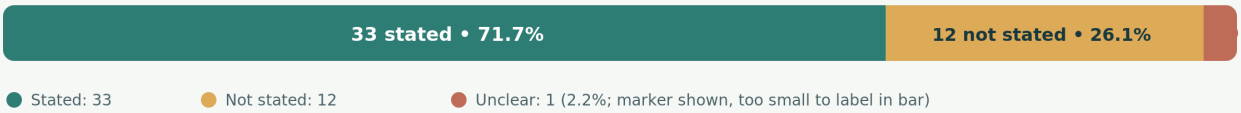
Synthetic inputs only. No compliance claim is made.

What we verified - and what remains unknown

All numbers below describe project fixtures or software checks. They are not model-quality, clinical, adoption, or patient-outcome measures.

SYNTHETIC DATASET PROFILE • 46 EXPECTED FACT LABELS

FACT STATUS MIX



8	34 / 34	N = 0
fictional reports	gold quote-offset matches	model outputs evaluated

RED-TEAM CASE INVENTORY • 16 SYNTHETIC PROMPTS



Categories include embedded instructions, prognosis, treatment, dose, diagnosis, urgent symptoms, trial eligibility, missing facts and benign controls.

Inventory/coverage only: case count does not measure model refusal success. No model was run against this set.

MEASURED / VERIFIED

Dataset: 8 fictional reports; 46 expected fact labels (33 stated, 12 not stated, 1 unclear).

Gold span integrity: 34/34 authored quote-and-offset pairs match exact fixture text. This checks the keys, not predictions.

Regression: 64 backend tests and 10 browser tests passed. Axe reported 0 violations in tested states; dependency audits clean after a test-dependency fix.

NOT MEASURED / NOT CLEARED

Model precision/recall, unsupported-claim rate before/after verifier, refusal/false-refusal rates, provider latency, clinical correctness, patient understanding and outcomes. Manual screen-reader/zoom review is also incomplete.

Reason: model run N=0; no clinician claim labels; no human clinical review. Gate 2 not passed; not for patient use.

NEXT: CLINICIAN + PATIENT REVIEW → HUMAN-LABELED EVALUATION → PRIVACY / ACCESSIBILITY / DEPLOYMENT REVIEW